KAIZEN MATHS

Lesson starter

Linear Algebra: Systems, Span, Basis and Transformations

Change of Basis and Linear Maps

Name _____

S1

Find the coordinates of the vector relative to the given basis.

For the basis B = $\{\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}-4\\1\end{bmatrix}\}$, find $[\begin{bmatrix}20\\4\end{bmatrix}]_B$.

S2

Construct the requested change of basis matrix.

For the basis B = $\{\begin{bmatrix}-3\\4\end{bmatrix}, \begin{bmatrix}-4\\-3\end{bmatrix}\}$, find the matrix that converts B-coordinates into standard coordinates.

S3

Use the basis images and linearity to find the image of the vector.

Given $T(\begin{bmatrix}-4\\-4\end{bmatrix}) = \begin{bmatrix}2\\2\end{bmatrix}$ and $T(\begin{bmatrix}3\\-3\end{bmatrix}) = \begin{bmatrix}-1\\4\end{bmatrix}$, find $T(\begin{bmatrix}-1\\-7\end{bmatrix})$.

S4

Use rank-nullity to interpret the transformation.

A linear map $T: R^3 \to R^2$ has rank 2. Use rank-nullity to find nullity(T) and decide whether T can be one-to-one.

Show your working

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Construct the requested change of basis matrix.

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KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Change of Basis and Linear Maps

Name _____

EA1

Find the coordinates of the vector relative to the given basis.

For the basis B = $\{\begin{bmatrix}-1\\-3\end{bmatrix}, \begin{bmatrix}-4\\-1\end{bmatrix}\}$, find $[\begin{bmatrix}11\\0\end{bmatrix}]_B$.

EA2

Construct the requested change of basis matrix.

For the basis B = $\{\begin{bmatrix}-3\\-3\end{bmatrix}, \begin{bmatrix}-1\\-3\end{bmatrix}\}$, find the matrix that converts B-coordinates into standard coordinates.

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Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Change of Basis and Linear Maps

Name _____

EB1

Use the basis images and linearity to find the image of the vector.

Given $T(\begin{bmatrix}2\\4\end{bmatrix}) = \begin{bmatrix}-1\\2\end{bmatrix}$ and $T(\begin{bmatrix}-1\\4\end{bmatrix}) = \begin{bmatrix}-1\\2\end{bmatrix}$, find $T(\begin{bmatrix}9\\0\end{bmatrix})$.

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Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Change of Basis and Linear Maps

Name _____

EA1

Find the coordinates of the vector relative to the given basis.

For the basis B = $\{\begin{bmatrix}-1\\-3\end{bmatrix}, \begin{bmatrix}-4\\-1\end{bmatrix}\}$, find $[\begin{bmatrix}11\\0\end{bmatrix}]_B$.

EA2

Construct the requested change of basis matrix.

For the basis B = $\{\begin{bmatrix}-3\\-3\end{bmatrix}, \begin{bmatrix}-1\\-3\end{bmatrix}\}$, find the matrix that converts B-coordinates into standard coordinates.

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Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Change of Basis and Linear Maps

Name _____

EB1

Use the basis images and linearity to find the image of the vector.

Given $T(\begin{bmatrix}2\\4\end{bmatrix}) = \begin{bmatrix}-1\\2\end{bmatrix}$ and $T(\begin{bmatrix}-1\\4\end{bmatrix}) = \begin{bmatrix}-1\\2\end{bmatrix}$, find $T(\begin{bmatrix}9\\0\end{bmatrix})$.

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Teacher answers & working

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S1

Find the coordinates of the vector relative to the given basis.

For the basis B = $\{\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}-4\\1\end{bmatrix}\}$, find $[\begin{bmatrix}20\\4\end{bmatrix}]_B$.

Answer:

$[\begin{bmatrix}20\\4\end{bmatrix}]_B = \begin{bmatrix}4\\-4\end{bmatrix}$

Working:

Find scalars a and b such that $\begin{bmatrix}20\\4\end{bmatrix} = a\begin{bmatrix}1\\2\end{bmatrix} + b\begin{bmatrix}-4\\1\end{bmatrix}$.

Equating components gives a − 4b = 20 and 2a + b = 4.

Solving gives a = 4 and b = −4.

Therefore $[\begin{bmatrix}20\\4\end{bmatrix}]_B = \begin{bmatrix}4\\-4\end{bmatrix}$.

S2

Construct the requested change of basis matrix.

For the basis B = $\{\begin{bmatrix}-3\\4\end{bmatrix}, \begin{bmatrix}-4\\-3\end{bmatrix}\}$, find the matrix that converts B-coordinates into standard coordinates.

Answer:

$P_B = \begin{bmatrix}-3 & -4\\4 & -3\end{bmatrix}$

Working:

A coordinate vector $\begin{bmatrix}a\\b\end{bmatrix}_B$ means $a\begin{bmatrix}-3\\4\end{bmatrix} + b\begin{bmatrix}-4\\-3\end{bmatrix}$.

The conversion matrix uses the basis vectors as its columns.

$P_B = \begin{bmatrix}-3 & -4\\4 & -3\end{bmatrix}$.

Check that $\det(P_B) = 25$, which is not zero, so B is a valid basis.

Teacher answers & working

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S3

Use the basis images and linearity to find the image of the vector.

Given $T\begin{bmatrix}-4\\-4\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$ and $T\begin{bmatrix}3\\-3\end{bmatrix} = \begin{bmatrix}-1\\4\end{bmatrix}$, find $T\begin{bmatrix}-1\\-7\end{bmatrix}$.

Answer:

$T\begin{bmatrix}-1\\-7\end{bmatrix} = \begin{bmatrix}1\\6\end{bmatrix}$

Working:

First express the input vector using the basis vectors.

$\begin{bmatrix}-1\\-7\end{bmatrix} = 1\begin{bmatrix}-4\\-4\end{bmatrix} + 1\begin{bmatrix}3\\-3\end{bmatrix}$.

Use linearity: $T(a\begin{bmatrix}-4\\-4\end{bmatrix} + b\begin{bmatrix}3\\-3\end{bmatrix}) = aT(\begin{bmatrix}-4\\-4\end{bmatrix}) + bT(\begin{bmatrix}3\\-3\end{bmatrix})$.

$T\begin{bmatrix}-1\\-7\end{bmatrix} = 1\begin{bmatrix}2\\2\end{bmatrix} + 1\begin{bmatrix}-1\\4\end{bmatrix}$.

$T\begin{bmatrix}-1\\-7\end{bmatrix} = \begin{bmatrix}1\\6\end{bmatrix}$.

S4

Use rank-nullity to interpret the transformation.

A linear map $T: R^3 \to R^2$ has rank 2. Use rank-nullity to find nullity(T) and decide whether T can be one-to-one.

Answer:

nullity(T) = 1. It cannot be one-to-one.

Working:

Rank-nullity states: rank(T) + nullity(T) = dimension of the domain.

2 + nullity(T) = 3.

nullity(T) = 3 − 2 = 1.

A non-zero nullity means there is at least one non-zero vector in the kernel, so T cannot be one-to-one.

Teacher answers & working

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EA1

Find the coordinates of the vector relative to the given basis.

For the basis B = $\{\begin{bmatrix}-1\\-3\end{bmatrix}, \begin{bmatrix}-4\\-1\end{bmatrix}\}$, find $[\begin{bmatrix}11\\0\end{bmatrix}]_B$.

Answer:

$[\begin{bmatrix}11\\0\end{bmatrix}]_B = \begin{bmatrix}1\\-3\end{bmatrix}$

Working:

Find scalars a and b such that $\begin{bmatrix}11\\0\end{bmatrix} = a\begin{bmatrix}-1\\-3\end{bmatrix} + b\begin{bmatrix}-4\\-1\end{bmatrix}$.

Equating components gives −a − 4b = 11 and −3a − b = 0.

Solving gives a = 1 and b = −3.

Therefore $[\begin{bmatrix}11\\0\end{bmatrix}]_B = \begin{bmatrix}1\\-3\end{bmatrix}$.

EA2

Construct the requested change of basis matrix.

For the basis B = $\{\begin{bmatrix}-3\\-3\end{bmatrix}, \begin{bmatrix}-1\\-3\end{bmatrix}\}$, find the matrix that converts B-coordinates into standard coordinates.

Answer:

$P_B = \begin{bmatrix}-3 & -1\\-3 & -3\end{bmatrix}$

Working:

A coordinate vector $\begin{bmatrix}a\\b\end{bmatrix}_B$ means $a\begin{bmatrix}-3\\-3\end{bmatrix} + b\begin{bmatrix}-1\\-3\end{bmatrix}$.

The conversion matrix uses the basis vectors as its columns.

$P_B = \begin{bmatrix}-3 & -1\\-3 & -3\end{bmatrix}$.

Check that $\det(P_B) = 6$, which is not zero, so B is a valid basis.

Teacher answers & working

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EB1

Use the basis images and linearity to find the image of the vector.

Given $T\begin{bmatrix}2\\4\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$ and $T\begin{bmatrix}-1\\4\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$, find $T\begin{bmatrix}9\\0\end{bmatrix}$.

Answer:

$T\begin{bmatrix}9\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$

Working:

First express the input vector using the basis vectors.

$\begin{bmatrix}9\\0\end{bmatrix} = 3\begin{bmatrix}2\\4\end{bmatrix} - 3\begin{bmatrix}-1\\4\end{bmatrix}$.

Use linearity: $T(a\begin{bmatrix}2\\4\end{bmatrix} + b\begin{bmatrix}-1\\4\end{bmatrix}) = aT(\begin{bmatrix}2\\4\end{bmatrix}) + bT(\begin{bmatrix}-1\\4\end{bmatrix})$.

$T\begin{bmatrix}9\\0\end{bmatrix} = 3\begin{bmatrix}-1\\2\end{bmatrix} - 3\begin{bmatrix}-1\\2\end{bmatrix}$.

$T\begin{bmatrix}9\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$.